

Assignment 1: Computer Architecture, Organization, and Performance Measurement

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Deadline: two weeks from the date of issue — there is no automatic late penalty; if you need more time, ask before the deadline and an extension will be granted (see the course policy in the syllabus).

Answer every question. Show all working for Numerical and Design questions — a bare final answer without the derivation earns partial credit only.

1.1 Conceptual

Q1. A new model of an existing x86 processor ships with a redesigned datapath that halves the CPI of most instructions but keeps the instruction set, registers, and addressing modes unchanged. A binary compiled years ago runs unchanged and measurably faster on the new model. Explain this observation using the distinction between *computer architecture* and *computer organization*: which of the two changed, which stayed fixed, and why old software continues to work. Your answer must use both consequences — compatibility and design freedom — from the lecture.

Q2. Two machines run the same program P . Machine A reports MIPS = 1000; Machine B reports MIPS = 800. A student concludes that A is faster. Explain why the MIPS numbers alone cannot support that conclusion, state the metric that *should* be used instead, and identify the one special condition under which the MIPS comparison would happen to be a safe guide.

1.2 Numerical

Q3. Program P executes 2×10^8 instructions. Machine A runs at $f = 2.5$ GHz with CPI = 1.8; Machine B runs at $f = 3.0$ GHz with CPI = 2.2. Compute the CPU time of both machines, state which is faster, and by what factor.

Q4. A program's total execution time is 200 s. A new floating-point unit makes the FP portion of the program $3\times$ faster. Compute the overall speedup, and separately compute the *best possible* overall speedup (i.e. the ceiling when the FP part is made infinitely fast).

1.3 Design

Q5. A program spends 25% of its time on load/store instructions. An engineer proposes a new memory system that makes loads/stores $8\times$ faster. (a) Compute the resulting overall speedup. (b) Compute the hard ceiling on overall speedup that this proposal can ever reach, no matter how

much faster loads/stores become. (c) The engineering target is a $2\times$ overall speedup — using your two numbers, argue whether investing further in the load/store path is the right place to spend design effort, and where the remaining gain must come from.

Q6. You are required to achieve a $4\times$ overall speedup on a workload by improving only its floating-point portion, which you may make *infinitely* fast. Using Amdahl's Law, determine the minimum fraction f of execution time that the FP portion must occupy for this goal to be achievable, and give the resulting speedup expression that confirms it.